

Technology Stack & Architecture Matrix

Date	19 September 2026
Team ID	
Project Name	ShopEZ — Apparel E-Commerce Application
Maximum Marks	4 Marks

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interactive single-page application handling client-side routing, state, and category/search filtering.	React.js, Vite, Tailwind CSS, React Router DOM
2	Application Logic — Frontend API Layer	A shared client service layer handling all HTTP communication with the backend, with the JWT automatically attached to each request.	Axios
3	Application Logic — Backend Services	Handles routing, request validation, and business logic (stock checks, order totals) across all resources.	Node.js, Express.js
4	Application Logic — Security	Authenticates users, hashes passwords, and issues/verifies signed session tokens; gates admin-only routes by role.	JSON Web Token (JWT), bcryptjs
5	Database	Document-oriented storage for users, products (with embedded variants), carts, orders, reviews, and site configuration.	MongoDB (NoSQL Document Store)
6	Object Data Mapping	Schema definition, validation, and query interface between the Express backend and MongoDB.	Mongoose ODM, MongoDB Atlas (cloud-hosted cluster)
7	Infrastructure / Deployment	Runtime environment for both frontend and backend during development.	Node.js runtime (v18+); currently run locally via npm run dev / nodemon — not yet deployed to a public host

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology Implementation
1	Open-Source Frameworks	Every layer of the stack — frontend, backend, and database — is built on freely available, widely-adopted open-source technology.	MERN Stack (MongoDB, Express, React, Node.js)
2	Security Implementation	Passwords are never stored in plain text; sessions are stateless and verified on every protected request.	bcrypt password hashing, server-signed JWTs, protect/adminOnly middleware
3	Scalable Architecture	A clean three-layer separation (routes/middleware, controllers, models) means new features are added without touching existing code paths.	MVC pattern via Express + Mongoose
4	Data Integrity	Variant-level stock and order-time snapshotting prevent overselling and keep historical orders accurate even after a product changes.	Mongoose schema validation, atomic stock decrement on order placement
5	Performance	Client-side routing avoids full page reloads; debounced search and lean queries keep the catalog responsive.	React Router SPA navigation, debounced search input, indexed MongoDB queries